

Engineering Physics: Quantum Mechanics Fundamentals

Unit 1: Quantum Foundations

1.1 Wave-Particle Duality and de Broglie Wavelength

In 1924, Louis de Broglie postulated that matter exhibits wave-particle duality. The de Broglie wavelength λ is given by:

$$\lambda = h / p = h / (m \cdot v)$$

where h is Planck's constant (6.626×10^{-34} J s), p is momentum, m is particle mass, and v is velocity.

For an electron accelerated through potential difference V , the kinetic energy is $E = q \cdot V = (1/2) \cdot m \cdot v^2$.

Therefore, the wavelength can be expressed as:

$$\lambda = h / \sqrt{2 \cdot m \cdot q \cdot V}$$

1.2 The Time-Dependent Schrödinger Equation

The central equation of non-relativistic quantum mechanics is the Time-Dependent Schrödinger Equation (TDSE):

$$i \cdot \hbar \cdot (\partial \psi / \partial t) = - (\hbar^2 / (2 \cdot m)) \cdot (\nabla^2 \psi) + V(r, t) \cdot \psi$$

where $\psi(r, t)$ is the complex wavefunction, $\hbar = h / (2 \cdot \pi)$ is the reduced Planck constant, and $V(r, t)$ is the potential energy.

The physical interpretation given by Max Born states that $|\psi(r, t)|^2$ represents the probability density of finding a particle at position r at time t .

$$P(r) = |\psi(r, t)|^2 = \psi^* \cdot \psi$$

The total probability integrated over all space must equal 1 (Normalization Condition):

$$\int |\psi(r, t)|^2 dV = 1$$

Unit 2: One-Dimensional Bound Systems

2.1 Particle in a 1D Infinite Potential Well

Consider a particle of mass m confined to a 1D box with potential:

$$V(x) = 0 \text{ for } 0 \leq x \leq L$$

$$V(x) = \text{infinity for } x < 0 \text{ or } x > L$$

Inside the well, the Time-Independent Schrödinger Equation (TISE) simplifies to:

$$-\left(\frac{\hbar^2}{2m}\right) \frac{d^2 \psi}{dx^2} = E \psi$$

$$\frac{d^2 \psi}{dx^2} + k^2 \psi = 0, \text{ where } k^2 = 2mE / \hbar^2$$

Applying boundary conditions $\psi(0) = 0$ and $\psi(L) = 0$ yields discrete energy eigenvalues:

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} = \frac{n^2 \pi^2 \hbar^2}{8mL^2}, \text{ for } n = 1, 2, 3, \dots$$

The normalized wavefunctions are:

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

Summary of Energy Levels:

Level $n = 1$ (Ground state): $E_1 = \frac{\hbar^2}{8mL^2}$, Zero-point energy

Level $n = 2$ (First excited): $E_2 = 4 \cdot E_1$

Level $n = 3$ (Second excited): $E_3 = 9 \cdot E_1$

2.2 Quantum Tunneling and Barrier Penetration

When a particle with energy E encounters a rectangular potential barrier of height V_0 where $E < V_0$, classical $\psi(x) \sim \exp(-\alpha x)$, where $\alpha = \sqrt{2m(V_0 - E)} / \hbar$

Transmission coefficient T is non-zero:

$$T \approx 16 \cdot \left(\frac{E}{V_0}\right) \cdot \left(1 - \frac{E}{V_0}\right) \cdot \exp(-2\alpha a)$$

where a is the barrier width. This phenomenon explains alpha decay in radioactive nuclei and scanning tunneling